

Slide	Notes
C O D E Lesson 11: Ownership, Ethics, and Creativity Problem Solving with AI	
Warm Up	Warm Up (5 mins)
Problem Solving with AI Ownership, Ethics, and Creativity – Warm Up ✔ Do This: Decide on whether the scenarios on the next slides feel real and fair or misleading and unfair Put your thumb up, down or in the middle based on your opinion	The Reality Check: Real and Fair or Misleading and Unfair? (5 mins) ■ Goal of this activity: To spark students' thinking about how AI-generated content can raise questions of fairness, originality, and trust — especially when it blurs the line between human and machine-made work. ■■ Do This: Introduce the warm-up by explaining that students will see real examples of how AI is used to create things like songs, videos, or writing. For each example, give students approximately 30 seconds to decide whether it feels real and fair or misleading and unfair. Then, have them show a thumbs up if it feels real and fair, a thumbs down if it feels misleading and unfair, or a sideways thumb if they're unsure or in the middle. Let students know you'll call on a few volunteers after each round to briefly share why they chose their position. Teaching Tip This warm-up is meant to spark curiosity and preview the ethical challenges they'll explore more deeply in the main activity — not to settle any debates yet. Keep the pace brisk: just 30 seconds for students to decide, then a quick reason from 1–2 volunteers. Don't let discussion go too deep here — remind students they'll have more time to dig into these ideas during the station activity.

The Reality Check

AI-generated news article

Thumbs Up: Feels real and fair

Thumbs Down: Feels misleading and unfair



■ ■ Do This: Read the scenario aloud: “AI-generated news article.” Give students about 30 seconds to decide whether this example feels real and fair or misleading and unfair. Then, they show a thumb up, down or sideways. Call on a few students to briefly explain their reasoning.

■ Discuss: Thumbs up if you think an AI-generated news article feels real and fair. Thumbs down if you think it feels misleading and unfair. Thumbs in the middle if you are unsure.

■ Discussion Goal: Students begin to question the trustworthiness and authorship of AI-generated content. This example invites discussion about whether AI-generated writing should be trusted, who’s accountable for errors, and whether it’s ethical to present AI-written news without clear labeling.

Teaching Tip

You might mention the real-world case of CNET, which published dozens of AI-written articles in 2023. Many contained factual errors or plagiarism and led to public backlash. Ask students, “If the article looks professional but it is wrong, who should be held responsible?”

The Reality Check

AI-generated audio recording based on someone’s actual voice

Thumbs Up: Feels real and fair

Thumbs Down: Feels misleading and unfair



■ ■ Do This: Read the scenario aloud: “AI-generated audio recording based on someone’s actual voice.” Give students about 30 seconds to decide whether this example feels real and fair or misleading and unfair. Then, they show a thumb up, down or sideways. Call on a few students to briefly explain their reasoning.

■ Discuss: Thumbs up if you think an AI-generated audio recording based on someone’s actual voice feels real and fair. Thumbs down if you think it feels misleading and unfair. Thumbs in the middle if you are unsure.

■ Discussion Goal: Help students recognize that AI-generated voice recordings raise serious questions about consent, impersonation, and control over identity. They should begin to consider whether it’s ethical to mimic someone’s voice — especially when used in entertainment or media — without permission.

Teaching Tip

You can reference the real 2023 example of an AI-generated song using the voices of Drake and The Weeknd that went viral — even though neither artist participated. Ask students, “If an AI uses someone’s voice without their permission, who benefits — and who’s harmed?”

The Reality Check

AI filter or airbrushed photo

Thumbs Up: Feels real and fair

Thumbs Down: Feels misleading and unfair



■ ■ Do This: Read the scenario aloud: “AI filter or airbrushed photo.” Give students about 30 seconds to decide whether this example feels real and fair or misleading and unfair. Then, they show a thumb up, down or sideways. Call on a few students to briefly explain their reasoning.

■ Discuss: Thumbs up if you think an AI filter or airbrushed photo feels real and fair. Thumbs down if you think it feels misleading and unfair. Thumbs in the middle if you are unsure.

■ Discussion Goal: Encourage students to think critically about how AI tools for filtering or retouching photos influence our perceptions of reality, beauty, and authenticity. They might surface ideas about consent (when someone’s image is altered), self-representation, or pressure to meet unrealistic standards. You can also connect this to broader questions of what counts as original or truthful in creative work.

 Discuss:

If AI content (like a deepfake video or audio) looks or sounds like a real person, but a real person didn't create it, who should get credit or control?

What makes an AI creation feel unfair to the original person or artist involved?



■ Discuss: Click through the animated slide to display the prompts.

If AI content (like a deepfake video or audio) looks or sounds like a real person, but a real person didn't create it, who should get credit or control?

What makes an AI creation feel unfair to the original person or artist involved?

■ Discussion Goal: Students start thinking critically about representation and ownership. By referencing deepfakes explicitly, the class can discuss what happens when AI convincingly imitates someone's voice or face — and how that differs from AI remixing someone's creative work. This sets up both the vocabulary (like attribution and copyright) and the upcoming station questions about credit, control, and fairness.

■■ Say: If we can't tell the difference between what AI makes and what people make, how should we handle credit, permission, and payment? That's exactly what we're going to explore in the next activity.

Lesson Objectives

By the end of this lesson, you will be able to . . .

- Justify how AI should be used in creative situations, considering ethical, practical, and societal values.
- Evaluate the role of AI as a collaborator in creative work, including how it can support or undermine the contributions of human creators.

■■ Do This: Introduce the lesson objectives.

Question of the Day

What are the ethical considerations of AI collaboration?

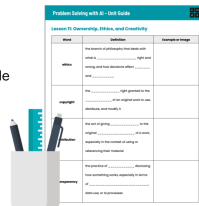
■■ Do This: Introduce the Question of the Day.

Activity (35 mins)

Activity**Unit Guide**

You should have:

- Problem Solving with AI Unit Guide
- pen or pencil



■■ Do This: Have students take out their Problem Solving with AI Unit Guide.

What is Ethics?

Ethics is a guide for responsible choices about emerging technologies, ensuring privacy, consent, and safety for everyone


[Unit Guide](#)

Vocabulary and Real World Examples (10 mins)

■ Goal of this activity: Introduce foundational concepts to help students evaluate AI-generated content legally and ethically and analyze AI copyright challenges through real-world disputes.

■■ Do This: Define ethics.

Teaching Tip

After defining ethics, ask students to think about how using someone else's work without asking might be considered an ethical issue — even if it's not illegal. Don't get into examples yet; just note that ethics often involve how people are treated, not just what rules exist.

What is Copyright?

Copyright is the legal right granted to the creator of an original work to use, distribute, and modify it.


[Unit Guide](#)

■■ Do This: Define copyright.

Teaching Tip

Let students know this is a legal concept, not just about giving credit — it's about who has the right to decide how a piece of work is used. You might find it helpful to tell students that if they create something, they decide whether others can copy it or profit from it. That legal control is what copyright protects.

What is Attribution?

Attribution refers to the act of giving credit to the original creator of a work, especially in the context of using or referencing their material.


[Unit Guide](#)

■■ Do This: Define attribution and emphasize that it doesn't give the creator control, but it does recognize their contribution. It's often seen as a basic form of respect in creative work.

What is Transparency?

Transparency is the practice of openly disclosing how something works, especially in terms of decision-making, data use, or AI processes.


[Unit Guide](#)

■■ Do This: Define transparency.

Teaching Tip

You might tell students, that transparency matters when people don't know if their work is being used, or how AI systems are making decisions. This helps students understand why lack of transparency can lead to ethical problems — especially if creators never know they were involved.



OpenAI has published documentation describing how it trains its models, but it doesn't always name specific datasets. Some creators argue that this lack of **transparency** makes it hard to know if their work was used—and without **attribution**, they may not get credit.

■■ Do This: Introduce the idea of transparency policies by explaining OpenAI shares general information about how it trains its models but does not always list exactly what data was used. Encourage students to think about how this may make some creators raise questions about a lack of transparency making it impossible to tell if their work was included without permission and attribution.

Teaching Tip

This is a good opportunity to reinforce the vocabulary words "transparency" and "attribution." Consider asking students how they would feel if their own artwork or writing were used to train an AI without them knowing or would it feel different if their name was somewhere in the training list, but the AI never directly quoted them or copied their style. However, don't spend too much time on these next three slides as the bulk of this activity should be devoted to student work time.

The European Union's proposed AI Act would require companies to say if **copyrighted** works were used to **train** their models—and give creators the right to opt out. It also sets **ethical** rules for different kinds of AI systems based on their risk level: the more risk, the stricter the rules.



■ ■ Do This: Introduce the EU AI Act as an example of how governments are starting to regulate AI. Point out that it requires companies to say if they used copyrighted content and gives creators the option to opt out. Use the pyramid graphic to explain how the Act assigns different rules based on an AI system's risk level, with higher-risk systems requiring stronger ethical protections.

In 2023, artists and authors sued AI companies, arguing their copyrighted works were scraped and used without permission. They believe this practice is not just illegal—but also unethical—because it harms their ability to earn a living.



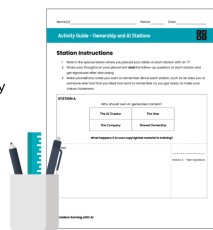
Image Credit: Creator: Devrim/Getty Images

■ ■ Do This: Emphasize that this example was one of the strongest reactions to AI training practices and provide the context that in 2023 artists and authors sued AI companies for using their copyrighted works without their permission. Encourage students to consider that this is not just a legal issue but an ethical one as well as it affects the livelihoods of these creators.

Ownership and AI

You should have:

- Ownership and AI Stations activity guide



Ownership and AI Stations (20 mins)

■ Goal of this activity: Students analyze real-world AI controversies to explore the ethical tradeoffs between innovation and creative ownership. They'll consider issues like consent, credit, and compensation while hearing multiple perspectives and forming their own opinions.

■ ■ Say: Now that we've looked at some real-world examples of how AI is used in creative work, let's explore the bigger question: How can we balance the benefits of AI with fairness to original creators? At each station, you'll see a real controversy and weigh in with your own perspective.

■ ■ Do This: Place students in groups of 2-3.

■ Distribute: Give each student a copy of the Ownership and AI Stations activity guide.

Do This:

Travel around the classroom to discuss the different scenarios:

1. Read the scenario and questions.
2. Answer both questions on your sticky notes, providing your reasoning.
3. Place your stickies on the appropriate location on the chart
4. Discuss your views with the group.

Station A Poster

Who should own AI-generated content?

The AI Creator	The User
The Company	Shared Ownership

What happens if AI uses copyrighted material in training?

■ Do This: Explain the Ownership and AI Stations and the steps for this activity. Direct students to complete the stations with their groups. They will have about 2-3 minutes per station.

■ Circulate: As students work, listen for nuanced reasoning and respectful disagreements. Encourage them to use terms like consent, credit, copyright, and attribution in their discussions.

■ Do This: Direct students to go back to the station they started at for about 1 minute to take a look at the sticky placement and read their peer's sticky note comments. Once time is up, direct students to the station they went to second and repeat. Remind students to write down any new ideas or reflections they have on their activity guide as they visit each station.

Teaching Tip

Timing: Before students rotate, give them a 10-second countdown to finish writing and posting. After students complete all stations, allow them to revisit their first and second stations to see new responses. (They don't need to revisit their third station — they just came from it!)

Use the gif: While going through the instructions for the stations, it may be helpful to either talk through the gif on the slide or demonstrate the flow to ensure students understand that they will be writing on and putting two sticky notes on each poster.

The first sticky note at each station will be used to place on a scale or on a chart after they write their reasons for their placement on that first sticky note

The second sticky note at each station will be used to write their answer to the second question and they will place this second sticky note under the question on the poster.

You can determine whether you would like students to put their names on their sticky notes to track participation. If you decide to have students add their names, it is suggested that they put their names on the back of the sticky notes so that students are more focused on the responses, rather than who wrote what.

Station A Poster

Who should **own** AI-generated content?

The AI Creator	The User
The Company	Shared Ownership

What happens if AI uses copyrighted material in training?

■ Do This: Direct students' attention to Station A.

■ Discuss: Have a quick discussion about the placement of sticky notes on the different quadrants. What do they notice? What do they wonder? Click through the animated slide to display the second question, and pick out a couple of sticky notes from the second question on the poster to help drive the discussion of ownership and copyright.

■ Discussion Goal: The primary goal of this discussion is for students to explore how AI-generated content raises questions about ownership and copyright. They should consider who should control or benefit from content created by AI — especially when it is trained on human-created works. Students may mention shared ownership, consent, and whether companies or users should have legal rights to something built on other people's creative work.

Teaching Tip

To keep the station debriefs tight, try to spend about 1-2 minutes per station's poster and focus the discussions on the second question.



Station B Poster



Should AI-generated art be considered “real art”?



Who should get credit—the AI, the programmer, or the person providing the prompt?



■ Do This: Direct students' attention to Station B.

■ Discuss: Have a quick discussion about the placement of sticky notes along the real to not real scale. What do they notice? What do they wonder? Click through the animated slide to display the second question, and pick out a couple of sticky notes from the second question on the poster to help drive the discussion of authorship and creativity.

■ Discussion Goal: The primary goal of this discussion is for students to reflect on how AI-generated art challenges definitions of authorship and creativity. They should consider attribution — who deserves credit — when a person provides a prompt, but the machine generates the final output. They may connect this to whether the AI itself, the programmer, or the user should be seen as the “creator,” and whether competitions should require disclosure or separate AI-assisted work from human work.



Station C Poster



Should musicians have **control** over AI trained on their music?

Yes	No

Who should be compensated when AI-generated music is sold?



■ Do This: Direct students' attention to Station C.

■ Discuss: Have a quick discussion about the placement of sticky notes on the Yes or No table. What do they notice? What do they wonder? Click through the animated slide to display the second question, and pick out a couple of sticky notes from the second question on the poster to help drive the discussion of compensation, consent, and transparency.

■ Discussion Goal: The primary goal of this discussion is for students to analyze how AI-generated music raises concerns about compensation, consent, and transparency. They should explore whether musicians should have control over how their work is used to train AI, and who deserves to be paid when that training leads to profitable outputs. Students may debate whether using someone's style or voice without permission violates ethical or legal standards, even if the AI output is new.

Self-Assessment

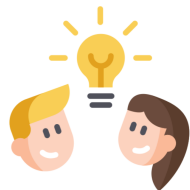
Lesson 11

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Do This:

- Go to Lesson 11, Level 1.
- Complete the sentence:

A value I think matters most when using AI with other people's work is ...



Check Your Understanding (5 mins)

■ Goal of this activity: To give students a chance to reflect on the ethical tradeoffs discussed at each station and articulate a value they think should guide how AI interacts with human-created content.

■ Transition: Navigate to Lesson 11, Level 1 on Code Studio in order to answer the free response reflection question.

Assessment Opportunity

Use this self-assessment to gauge how well students understand the ethical tradeoffs involved in AI development. As students respond, look for whether they:

Identify a clear value (e.g., attribution, transparency, consent) that reflects an ethical concern

Reference or imply key terms like attribution, transparency, compensation, or ownership

Demonstrate awareness of both benefits and risks of using human-created work in AI systems

Show consideration for who benefits and who may be left out when AI is trained on others' work

If many responses are vague or one-sided (e.g., “AI should always be allowed to use whatever data it wants”), consider doing one or more of the following:

Revisit one of the case studies from earlier in the lesson and discuss: “What might this creator say about that value?”

Have students revisit the vocabulary terms and identify which ones apply to their response

Use a class discussion to explore a range of student responses and ask students to challenge or build on one another's reasoning

Wrap Up	Wrap Up (5 mins)
Problem Solving with AI Ownership, Ethics, and Creativity – Wrap Up Today, you learned about . . . • How AI should be used in creative situations, considering ethical, practical, and societal values. • The role of AI as a collaborator in creative work, including how it can support or undermine the contributions of human creators.	■ ■ Do This: Review the lesson objectives.
Problem Solving with AI Ownership, Ethics, and Creativity – Wrap Up Question of the Day What are the ethical considerations of AI collaboration? Unit Guide	■ Goal of this activity: Students synthesize their learning and reflect on the ethical considerations of AI collaboration. ■ ■ Do This: Have students answer the Question of the Day in their Unit Guide and then share with their peers using a Think-Write-Share protocol: Students list ethical considerations in their activity guides Students share at one ethical consideration with a partner or small group. A few students share insights with the class.
Problem Solving with AI Ownership, Ethics, and Creativity – Wrap Up Key Vocabulary • ethics: the branch of philosophy that deals with what is morally right and wrong, and how decisions affect people and society • copyright: the legal right granted to the creator of an original work to use, distribute, and modify it	■ ■ Do This: Review the Key Vocabulary.
Problem Solving with AI Ownership, Ethics, and Creativity – Wrap Up Key Vocabulary • attribution: the act of giving credit to the original creator of a work, especially in the context of using or referencing their material • transparency: the practice of openly disclosing how something works, especially in terms of decision-making, data use, or AI processes	■ ■ Do This: Review the Key Vocabulary.